

# Chapter 72 — Evaluation and Management of Erectile Dysfunction

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## HISTORICAL PERSPECTIVE

- Sexual medicine is an **authentic, evidence-based discipline** resting on basic science, epidemiology, clinical investigation and health services research; the diagnosis and management of **erectile dysfunction (ED)** is its central element.
- **American Urological Association (AUA) definition:** the consistent or recurrent inability to attain and/or maintain penile erection **sufficient for sexual satisfaction**.
- Evolution: **1970s psychoanalysis, sex therapy, aphrodisiacs, herbal and hormonal remedies → contemporary structured, balanced, evidence-based evaluation and intervention**.
- Management principles rest on **ethics, quality, safety and cost-effectiveness**, codified by consensus bodies: the **AUA ED guideline and Core Curriculum**, and the **International Consultations on Sexual Medicine (ICSM)**, cosponsored by the World Health Organization (WHO), International Consultation on Urological Diseases (ICUD), Société Internationale d'Urologie (SIU) and International Society for Sexual Medicine (ISSM).

**TABLE 72.1 EVOLUTION IN THE MANAGEMENT OF ERECTILE DYSFUNCTION**

| Era          | Diagnostics                                                                                                                           | Treatments                                                                                         | Guides                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Pre-1970     | Psychosexual history                                                                                                                  | Psychosexual therapy; herbal supplements                                                           | Masters and Johnson                                                                                                  |
| 1970s        | Medical and psychosexual history; nocturnal penile tumescence testing                                                                 | <b>Penile prosthesis surgery; penile revascularisation</b>                                         | International Conferences on Corpus Cavernosum Revascularization                                                     |
| 1980s        | Physical examination; endocrine evaluation; penile duplex ultrasonography; dynamic infusion cavernosometry and cavernosography (DICC) | Oral medications; <b>intracavernous pharmacotherapy</b> ; vacuum device therapy                    | Goal-directed management                                                                                             |
| 1990s        | Combined intracavernous injection and stimulation                                                                                     | Intraurethral pharmacotherapy; <b>oral phosphodiesterase type 5 therapy</b>                        | National Institutes of Health (NIH) Consensus Statement; Process of Care Model                                       |
| 2000–present | Biomarkers of vascular health; neuroimaging                                                                                           | Gene, stem cell, tissue engineering, shock wave, hyperbaric oxygen — <b>all queried (unproven)</b> | ICUD algorithms (patient-centred); AUA guidelines (evidence-based); European Association of Urology (EAU) guidelines |

- **What the table tests:** the sequence of therapeutic eras. **Surgery (prosthesis, revascularisation) came first** (1970s), then local pharmacotherapy (1980s), then oral therapy (1990s) — the field moved from most to least invasive, not the reverse.

- **Trap:** every 2000-era treatment carries a query — regenerative therapies are investigational, not established.

**Concept** — *The history of ED management is a migration from subjective to objective diagnosis and from invasive-first to least-invasive-first treatment, ending in guideline-governed, patient-centred care.*

## PUBLIC HEALTH SIGNIFICANCE

- **NIH Consensus Statement (1992):** inability to attain and/or maintain erection sufficient for **satisfactory sexual performance**; later opinion accepts **sufficiency for sexual satisfaction** as the valid purpose.
- ED is a condition of **major health significance**: its management affects the individual, the partner and society — physical and mental wellness, concurrent disease management and **socioeconomic burden**.

## Epidemiology

- ED is the **most thoroughly studied sexual dysfunction**; population-generalisable epidemiology informs its nature, cause and prognosis.
- **Overall prevalence** in adult men worldwide 10–20%, most studies **closer to 20%**.
- **Age gradient:** under 40 years 1–10%; 40–49 up to 15%; 50–59 up to 30%; 60–69 up to 40%; 70–90 years 50–100%.
- Among men seeking first help for new-onset ED, **one in four is younger than 40**, and nearly half of these have severe ED.
- **Global burden:** over 152 million men in 1995, projected to **≈322 million by 2025**, a trend that holds across race, ethnicity and region.
- **Comorbidities raise prevalence:** type 2 diabetes mellitus (DM), obesity, cardiovascular disease (CVD), hypertension, dyslipidaemia, depression, prostate disease/benign prostatic hyperplasia (BPH) — sharing **endothelial dysfunction, arterial occlusion and systemic inflammation**.
- **Novel disease associations** (same mechanisms presumed): epilepsy, sensorineural hearing loss, open-angle glaucoma, urinary calculi, psoriasis, atopic dermatitis, chronic periodontitis, viral hepatitis, varicocele, gastric ulcer, ankylosing spondylitis, gout, inflammatory bowel disease/ulcerative colitis, interstitial cystitis, Parkinson disease, obstructive sleep apnoea, premature ejaculation.
- **Coronavirus disease 2019 (COVID-19):** ED diagnoses rose during the pandemic; prior infection carries a **markedly higher (nearly sixfold) risk** of ED, which **persists after recovery**; mechanism unclear — multiorgan vasculogenic, neurogenic and endocrine effects theorised.
- **Opioid use ≈ doubles ED risk**, more so in men **under 50** → screen for opioid use at initial assessment.
- **Incidence (prospective studies):** crude incidence **25.9 cases/1000 man-years** in men aged 40–69 (Massachusetts Male Aging Study cohort); incident ED 57% at 5 years and 65% at 7 years in men ≥55 (Prostate Cancer Prevention Trial placebo arm).
- **Predictors of incident ED:** age, lower education, DM, CVD, hypertension, **cigarette and cigar smoking, passive smoke exposure**, overweight.
- **ED predicts CVD** (the reverse direction):
  - **Prostate Cancer Prevention Trial (PCPT) placebo arm:** ED is a sentinel of cardiovascular events **comparable with current smoking or a family history of myocardial infarction**; men with ED were **45% more likely to have a cardiac event at 5 years**.
  - Community cohort: ED confers substantially higher coronary artery disease risk at 10 years.
  - **Massachusetts Male Aging Study (MMAS)**, 15-year follow-up: ED is associated with all-cause and CVD mortality, **comparable to raised body mass index (BMI), DM and hypertension**.
  - **Younger age at ED onset heightens the cardiovascular risk.**

- ED severity tracks overall health (Charlson Comorbidity Index) independent of cardiovascular (CV) risk factors.
- **Meta-analyses of cohorts** confirm a modest but consistent independent excess risk for CVD events, coronary heart disease, stroke/cerebrovascular events, myocardial infarction, CV mortality and all-cause mortality — largest for cerebrovascular and CVD events, smallest for all-cause mortality.
- **Cancer:** Taiwan Longitudinal Health Insurance Database — ED linked to higher adjusted cancer risk over 5 years.
- ED is a **clinical barometer of overall male health**; its management bears on prevention, health promotion and survival.

## Health Policy

- ED reduces **quality of life (QOL)** and occupational productivity and increases health care use; service use is rising with effective first-line therapy, awareness and acceptance.
- United States outpatient ED management (excluding drugs) was the **ninth most costly urological diagnosis** in 2000, having roughly doubled since 1994; government drug expenditure projected in the billions over a decade.
- Some urge weighing medical necessity and cost in insurance coverage; evidence supports **rational coverage**, and therapy is actually **underused relative to prevalence**. ED is **not a frivolous indication**.

### KEY POINTS: EPIDEMIOLOGY AND HEALTH POLICY

- Approximately **20% of adult men** worldwide experience ED.
- Risk factors: increasing age and comorbidity — DM, obesity, CVD, hypertension, depression, prostate disease.
- Outcomes research shows ED merits intervention for QOL and for health and life preservation.
- ED ranks among the **top 10 most costly urological diagnoses** in the United States.

**Concept** — Because ED and vascular disease share **endothelial dysfunction**, the risk association runs in both directions: comorbidity predicts ED, and ED predicts cardiovascular events and death — which is why ED is treated as a sentinel, not a lifestyle complaint.

## MANAGEMENT PRINCIPLES

- ED management differs from most urological disease: **diagnosis rests on the subjective complaint**, extensive testing is generally unnecessary, and **first-line oral therapy** is easily prescribed and frequently successful.
- **Direct-to-consumer (DTC) services** are rapidly shifting this paradigm.
- Despite apparent simplicity, management is a **structured process** of direct evaluation and validation.

## Early Detection

- Many men with ED carry adverse conditions and lifestyle factors (DM, CVD, prostate disease, overweight, smoking, inactivity) → an **increased global cardiometabolic risk profile**.
- Men with identifiable risk factors have or will develop ED → **clinical screening of these men** creates opportunities to diagnose and treat.

### TABLE 72.2 MAJOR ERECTILE DYSFUNCTION RISK FACTORS

Multivariate-adjusted associations; data from Francis et al., 2007, and Selvin et al., 2007

| Strength of adjusted association     | Factors                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Strongest</b>                     | <b>Antidepressant use</b> (highest of all), then <b>antihypertensive use</b>                                                |
| <b>Strong</b>                        | <b>Diabetes mellitus</b> (strongest disease association); obstructive urinary symptoms                                      |
| <b>Moderate</b>                      | Hypertension; benign prostate enlargement; current cigarette smoking; BMI $\geq 30$ kg/m <sup>2</sup> ; physical inactivity |
| <b>Weak to null after adjustment</b> | <b>Cardiovascular disease; hypercholesterolaemia</b>                                                                        |

- **Pattern:** drug exposures outrank every disease, and DM outranks every other disease.
- **Trap:** CVD and hypercholesterolaemia, though central to the pathophysiology, carry almost no independent odds once confounders are adjusted — their effect is absorbed by DM, hypertension and drugs.
- **Genotype influences ED risk:** molecular and genetic mechanisms underlie the phenotype; genetically determined biomarkers may define risk and responsiveness to specific therapy (**precision medicine**). Polymorphisms in pulmonary hypertension already predict benefit from **phosphodiesterase type 5 inhibitors (PDE5Is)**; genes linking ED and CV risk could target screening.
- **Medication-associated ED** accounts for **up to 25% of presentations**; commonest culprits are **thiazide diuretics,  $\beta$ -adrenoceptor antagonists and selective serotonin reuptake inhibitors (SSRIs)**.
- Drugs may also impair **desire, arousal and orgasm**, secondarily hampering erection.
- **Criteria for causal attribution** to a drug: increased prevalence versus placebo after stratification for known risk factors (or versus a therapeutically equivalent drug) **and** an experimentally established physiological mechanism.

TABLE 72.3 DRUGS ASSOCIATED WITH ERECTILE DYSFUNCTION

| Class             | Specific agents                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| Antihypertensives | Thiazide diuretics; <b>nonselective</b> $\beta$ -blockers                            |
| Antidepressants   | Tricyclics; SSRIs                                                                    |
| Antipsychotics    | Phenothiazines                                                                       |
| Antiandrogens     | Nonsteroidal (flutamide); steroidal; luteinising hormone-releasing hormone analogues |
| Antiulcer drugs   | Histamine H <sub>2</sub> -receptor antagonists ( <b>cimetidine</b> )                 |
| Cytotoxic agents  | Cyclophosphamide; methotrexate                                                       |
| Opiates           | Morphine                                                                             |

- **What the table is for:** the medication history — every class here has a switch option or a dose strategy (see Medication Change).
- **Trap:** the  $\beta$ -blocker named is the **nonselective** type; the H<sub>2</sub> antagonist named is **cimetidine** specifically (antiandrogenic).

### Shared Decision Making and Treatment Planning

- The plan varies for each patient and couple; an **informed decision-making process** follows a balanced, clinician-led discussion of **all options, medical and nonmedical**, with advantages, disadvantages and individually weighted risks and benefits.
- The patient may select a preferred option **without following a prescribed succession** of therapies, and may **defer treatment altogether**; the choice is pursued within the boundaries of safety.
- Derived from **goal-directed management** and the patient-centred framework.

- Meta-analysis of randomised trials: shared decision making yields **more knowledgeable patients** and **better congruence between chosen therapy and values**.
- The AUA ED guideline panel (2018) endorses shared decision making as the state of the art.

### Figure 72.1 — Algorithm for shared decision making and treatment planning (AUA)

- **Counsel male and partner** on: the value of psychosocial/relationship support from a trained professional; the importance of **lifestyle change (weight loss, exercise, smoking cessation)** for erectile function and overall health; the benefits and risks/burdens of **all non-contraindicated treatments** → **identify treatment by shared decision making** on the values and priorities of male and partner: PDE5I, vacuum devices, intraurethral (IU) alprostadil, intracavernosal injections (ICIs), penile prosthesis surgery — **offered in parallel, not ranked** → **assess outcomes, adverse events (AEs) and satisfaction** of male and partner → if inadequate efficacy, unacceptable AEs or insufficient satisfaction: adjust dose (PDE5I, IU alprostadil, ICI), revisit instructions, revisit values and priorities, refer to a mental health professional for psychosocial or relationship barriers, or consider an alternate treatment.
- **Testosterone deficiency** (symptoms and signs plus **total testosterone <300 ng/dL**): counsel that restoring testosterone is likely to **increase the efficacy of ED treatments other than prosthesis surgery**.

### Step-Care Approach

- **Process of Care Model (1999)**: a stepwise methodology combining processes, actions and outcomes; its algorithm considers needs and preferences (goal-directed) but ranks therapies by **ease of administration, reversibility, invasiveness and cost** into **first-, second- and third-line** interventions, from lifestyle modification to surgery.
- Endorsed by later panels, which placed **education and counselling with medical therapy** as initial management.
- Contemporary management **emphasises shared decision making over step-care**.

### COMPARISON: STEP-CARE VERSUS SHARED DECISION MAKING

|                    | Step-care (Process of Care, 1999)                         | Shared decision making (AUA, 2018)                       |
|--------------------|-----------------------------------------------------------|----------------------------------------------------------|
| Ordering principle | Ease, <b>reversibility, invasiveness</b> , cost           | <b>Patient and partner values</b> and priorities         |
| Sequence           | <b>First → second → third line</b> , lifestyle to surgery | <b>All options in parallel</b> ; any may be chosen first |
| Deferral           | Implicitly progressive                                    | Explicitly permitted                                     |
| Status             | Historical, still commonly used                           | <b>Current state of the art</b>                          |

### Role of Partner Interview

- The partner interview is **critical**: it alters diagnosis and treatment in a substantial proportion of cases (over half in some series).
- The partner provides a **new perspective** on sexual issues, insight into relationship quality, and his or her own role in the dysfunction; partner attitude affects **initiation of and adherence to therapy**.
- **Partners are affected**: female partners of men with ED more often have sexual dysfunction or cease sexual activity.
- Recommended: **additional office visits with the partner**, during which the patient conveys educational information.

### Cardiac Risk Assessment

- **Princeton Consensus Conferences** (three, since the early 2000s): **all men with ED, even asymptomatic, are regarded as having potential CVD risk**.

- ED **improves the sensitivity of screening for asymptomatic CVD in men with DM** and raises risk of CVD, coronary artery disease and stroke.
- Low-risk men still receive **minimum CVD management**: lifestyle counselling (activity, weight) with regular primary-care monitoring. A more comprehensive approach adds CV risk reduction and **confirmation of exercise tolerance** after noninvasive assessment.

#### PRINCETON CONSENSUS RISK STRATIFICATION (FIGURE 72.2, SECOND PRINCETON PANEL)

| Stratum       | Features                                                                                                                                       | Action                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| High          | Unstable or refractory angina; recent myocardial infarction; certain arrhythmias; <b>uncontrolled hypertension</b>                             | <b>Defer sexual activity</b> and any ED therapy until stabilised; cardiology referral, stress testing, risk reduction |
| Indeterminate | Not clearly high or low                                                                                                                        | Cardiovascular assessment and <b>restratification</b>                                                                 |
| Low           | —                                                                                                                                              | Initiate or resume sexual activity, or treat the sexual dysfunction                                                   |
| All           | Sexual inquiry → clinical evaluation → <b>risk-factor and coronary heart disease evaluation, treatment and follow-up for every man with ED</b> |                                                                                                                       |

#### Specialist Referral

- Effective oral therapy lets primary practitioners manage most ED; referral is for **complicated or atypical presentations** beyond nonspecialist practice.
- At referral, inform the patient (and partner) of the **rationale, costs, risks and possible outcomes** — patient-centred decision making.

#### COMPARISON: REFERRAL INDICATIONS AND CONSULTANT

| Indication                                          | Example                                         | Consultant                     |
|-----------------------------------------------------|-------------------------------------------------|--------------------------------|
| Failure of initial treatment                        | —                                               | Urologist                      |
| <b>Young patient with pelvic or perineal trauma</b> | —                                               | Urologist                      |
| Significant penile deformity                        | Peyronie disease, congenital chordee            | Urologist                      |
| Medicolegal                                         | Workers' compensation claims                    | Urologist                      |
| Complicated endocrinopathy                          | Secondary hypogonadism, pituitary adenoma       | Endocrinologist                |
| Complicated psychiatric/psychosexual disorder       | Refractory depression, hypoactive sexual desire | Psychiatrist                   |
| Vascular or neurosurgical lesion                    | Aortic aneurysm; lumbosacral disc disease       | Vascular surgeon; neurosurgeon |

#### Direct-to-Consumer Platforms

- DTC platforms flourished through **marketing, disease normalisation and easy access**, especially among younger men.
- Caution: **no appropriate medical evaluation** — questionnaire-only screening and **nonadherence to AUA guidelines**; DTC clinics offer **unproven therapies**, and few providers are urologists or even physicians.
- Hence the importance of **direct evaluation by a urologist**, in person or by appropriately selected telemedicine.

#### Follow-Up Care

- Essential: ensures continued success — **discontinuation is high when patients are not reassessed**.

- Also reassesses medical and psychosocial conditions, need for **dose titration or substitution**, adverse effects and drug interactions, and offers education for patient and partner.

**Concept** — ED management is deliberately **patient-driven in its choices but fixed in its safety gates**: the treatment is chosen by values, while cardiac stratification, referral triggers and follow-up are non-negotiable.

## DIAGNOSTIC EVALUATION

- The cornerstone: **detailed history (patient and partner), physical examination and proper laboratory tests**.
- Diagnosis rests on the man's **report of consistent inability** to attain and maintain an erection sufficient for satisfactory intercourse.
- The original NIH definition set no duration; later statements require a **minimum of 3 months**, except in trauma or surgically induced ED.

**FIGURE 72.3 — ICSM DIAGNOSTIC ALGORITHM FOR ED**

| Component of basic evaluation | Finding                                                                       | Action                                                                                                |
|-------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Medical history               | Reversible risk factors; <b>unstable CV condition; pelvic/perineal trauma</b> | Review and manage; further CV work-up; <b>specific imaging and vascular testing in young patients</b> |
| Sexual function assessment    | Problem with orgasm, ejaculation, genital pain or libido                      | Further evaluation of other sexual dysfunctions                                                       |
| Psychosocial assessment       | Psychosocial distress, including partner conflict                             | In-depth psychosocial evaluation and management                                                       |
| Focused physical examination  | Penile abnormality (Peyronie); prostatic disease; signs of hypogonadism       | Evaluate further and manage                                                                           |
| Basic laboratory tests        | Undiagnosed diabetes, hyperlipidaemia, <b>low testosterone</b>                | Further endocrine evaluation if needed                                                                |

*Outcome: findings support initiating treatment (review and discuss options) or necessitate further evaluation, including optional/specialised testing.*

## Sexual, Medical and Psychosocial History

- Brief checklists or questionnaires may open the evaluation, but a **detailed interview is routine**.
- Sexual history taken with **utmost sensitivity** and cultural competence; **private, confidential and nonjudgemental**.
- **Do not assume a monogamous, heterosexual relationship**.
- With the patient's approval, the **partner interview** corroborates history and confirms mutual goals.

## Sexual History

- The **central component**: confirms the complaint and delineates **onset, duration, conditions, severity and cause**.
- Conditions favouring erection: stimuli used, **erections on awakening**, self-stimulation. Conditions hindering: **performance anxiety**, inability with a designated partner, motivational factors.
- Other issues: partner availability, interest and health; medical events at onset; previous treatment attempts.

## COMPARISON: CLINICAL SEVERITY OF ED

| Severity | Ability to attain/maintain erection | Satisfactory performance |
|----------|-------------------------------------|--------------------------|
| Mild     | Minimally decreased                 | Intermittent             |

| Severity        | Ability to attain/maintain erection | Satisfactory performance |
|-----------------|-------------------------------------|--------------------------|
| Moderate        | Moderately decreased                | Infrequent               |
| Severe/complete | Substantially decreased             | Rare or absent           |

- **Cause** is categorised as **psychogenic** (psychological or interpersonal determinant), **organic** (endocrine, neurological or cardiovascular) or **mixed**. Full dichotomy is often impossible, but a predominant basis guides therapy.
- Establish whether ED is **primary or secondary** to another phase of the sexual response cycle (desire, ejaculation, orgasm), and whether reduced arousal **preceded or followed** the ED.

**TABLE 72.4 CLASSIFICATION OF ERECTILE DYSFUNCTION***Modified from Ralph and McNicholas, 2000 (UK management guidelines)*

| Psychogenic              | Organic                         |
|--------------------------|---------------------------------|
| Sudden onset             | Gradual onset                   |
| Complete immediate loss  | Incremental progression         |
| Situational dysfunction  | Global dysfunction              |
| Waking erections present | Waking erections poor or absent |

- **What the table tests:** the history alone can suggest aetiology. **Preserved waking/nocturnal erections are the single strongest pointer to psychogenic ED**, because they prove an intact neurovascular axis.

**Medical History**

- Identifies predictors, risk factors and comorbidities; its payoff is **reversible or treatable factors** whose correction may improve erections, and guidance on choice of therapy.
- Associated conditions: **disease states** (type 2 DM, CVD, hypertension, dyslipidaemia, neurological disease, hypogonadism, thyroid disorders); **trauma** (spinal cord injury, pelvic surgery or radiation, sexual injury); **medications and recreational substances**.
- Record age; register **depression, anxiety and anger** — their relationship with ED is **bidirectional**.

**Psychosocial History**

- Best sexual performance requires **wellness of mind and body together**; ascertain mental health problems, emotional stressors and relationship difficulties, past and present.
- Also occupational status, financial security, family life and social support.

**Physical Examination**

- **Highly recommended**; complements history and may reveal the cause. Comprises anthropometrics (height, weight, **waist circumference**), body habitus (secondary sexual characteristics), and CV, neurological and genital examination with **focus on the external genitalia**.
- **Bulbocavernosus reflex**: squeezing the glans → contraction of bulbocavernosus felt by a finger in the anus.

**COMPARISON: EXAMINATION FINDINGS AND IMPLIED AETIOLOGY**

| Finding                                                                                                                 | Implied basis                                                         |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Kallmann or Klinefelter habitus; gynaecomastia; poor masculinisation; abnormal testicular position, size or consistency | <b>Endocrine</b> (hypogonadism)                                       |
| Obesity; raised blood pressure; abnormal femoral or pedal pulses                                                        | <b>Vascular</b>                                                       |
| Abnormal genital/perineal sensation; <b>abnormal bulbocavernosus reflex</b>                                             | <b>Neurogenic</b> (peripheral neuropathy, neurological disease or DM) |
| Micropenis; congenital chordee; Peyronie plaques                                                                        | <b>Structural</b> impediment                                          |

## Questionnaires and Sexual Function Symptom Scores

- Self-administered questionnaires are **extremely useful adjuncts** that concur with self-report in establishing diagnosis.
- Early instruments were long — **Derogatis Sexual Function Inventory (245 items)**, **Golombok Rust Inventory of Sexual Satisfaction (GRISS; 28 items)** — aimed at separating psychogenic from organic ED or assessing the couple; later instruments captured trial end points; current emphasis is on **brief, practical self-report measures of presence, severity and treatment response**.

### COMPARISON: PRINCIPAL SEXUAL FUNCTION INSTRUMENTS

| Instrument                                                            | Structure and use                                                                                                                                        |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>International Index of Erectile Function (IIEF)</b>                | <b>15 items, 5 domains:</b> erectile function, orgasmic function, sexual desire, intercourse satisfaction, overall satisfaction; <b>most widely used</b> |
| <b>IIEF-5 / Sexual Health Inventory for Men (SHIM)</b>                | Abridged <b>5-item, erection-focused</b> version for routine practice; scores <b>severity</b>                                                            |
| Brief Male Sexual Function Inventory                                  | Brief self-report                                                                                                                                        |
| Center for Marital and Sexual Health Sexual Functioning Questionnaire | Brief self-report                                                                                                                                        |
| Changes in Sexual Functioning Questionnaire                           | Brief self-report                                                                                                                                        |
| Erectile Dysfunction Inventory of Treatment Satisfaction (EDITS)      | <b>Treatment satisfaction</b>                                                                                                                            |
| Male Sexual Health Questionnaire                                      | Desire, erection, ejaculation, satisfaction; clinical and research                                                                                       |
| Sexual Experience Questionnaire                                       | Health-related QOL: erection, individual and couple satisfaction                                                                                         |
| Visual-Scale Questionnaire                                            | Quick screening tool                                                                                                                                     |

### COMPARISON: IIEF-5 SEVERITY BANDS

| Score    | 5–7           | 8–11     | 12–16            | 17–21 | 22–25        |
|----------|---------------|----------|------------------|-------|--------------|
| Category | <b>Severe</b> | Moderate | Mild to moderate | Mild  | <b>No ED</b> |

- **Limitation:** questionnaires **do not distinguish cause** and may not reflect objective severity; further testing by system (vascular, neurological, endocrine) may be needed for corrective action.

## Cardiovascular Risk Assessment Tools

- CVD-risk prediction models — **Framingham Risk Score** or a global risk score (family history of coronary heart disease, BMI, metabolic biomarkers) — are a powerful first step in any man evaluated for ED.
- **Coronary artery calcium score** correlates with ED severity; a possible future prognostic marker pending validation.

## Laboratory Tests

- Part of systematic evaluation: confirm associated conditions and uncover **treatable or undetected disease**. Standard panel, then tailoring.
- **Recommended panel:** serum chemistries, **fasting glucose or glycated haemoglobin (HbA1c)**, full blood count, **lipid profile**, **morning serum total testosterone (T)**.
- Low total T → measure **free (or bioavailable) T** and **luteinising hormone (LH)**; prolactin may be added; thyroid tests at the clinician's discretion.
- **Prostate-specific antigen (PSA)** as needed if prostate pathology is suspected (which exogenous T could promote).

- Urine dipstick **glycosuria** suggests DM. Proposed serum surrogates and signalling biomarkers are **not established**.

**Concept** — The basic evaluation is designed to find **the disease behind the ED** — diabetes, dyslipidaemia, hypogonadism, cardiovascular risk — not to prove the ED, whose diagnosis rests on the man's own report.

## SPECIALIZED EVALUATION AND TESTING

- Aim: **diagnostic precision** to direct therapy. Technologies can define vasculogenic, neurogenic, endocrine or psychogenic causation, yet **treatment can often be planned without extensive testing**; testing is applied mainly by specialists in complex presentations.

**TABLE 72.5 EVIDENCE-BASED TESTS FOR ORGANIC ERECTILE DYSFUNCTION AND RECOMMENDATIONS**

Rosen et al., 2004 (ICSM); grading after Harbour and Miller, 2001

| Grade | Vascular                                                                                                               | Audiovisual sexual stimulation (AVSS)                                                   | Neurophysiological                                                                                                          |
|-------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| A     | No test reaches grade A                                                                                                |                                                                                         |                                                                                                                             |
| B     | DICC; ICI pharmacotesting; ICI with colour duplex ultrasound                                                           | —                                                                                       | Nocturnal penile tumescence and rigidity (NPTR); bulbocavernosus reflex latency                                             |
| C     | Arteriography                                                                                                          | Alone or with vascular testing; with or without pharmacological stimulation (oral, ICI) | Biothesiometry; dorsal nerve conduction velocity; corpus cavernosum electromyography (CC-EMG)                               |
| D     | Computed tomography angiography; magnetic resonance imaging (MRI); infrared spectrophotometry; radioisotope penography | —                                                                                       | Erectiometer/rigidometer; plethysmography/ electrobioimpedance; brain MRI or positron emission tomography (PET) during AVSS |

Grades: **A** — at least one meta-analysis, systematic review or low-bias randomised trial directly applicable; **B** — high-quality systematic reviews of case-control or cohort studies, consistent; **C** — well-conducted case-control or cohort studies, low confounding, moderate probability of causality; **D** — nonanalytic studies (case reports, series, expert opinion).

- **What the table is for:** which specialised tests are defensible. **Only five tests reach grade B** — the three ICI-based vascular tests, NPTR and bulbocavernosus reflex latency.
- **Trap:** arteriography, though the anatomical reference for revascularisation, is only grade C; imaging-only modalities (computed tomography, MRI, penography) are grade D.

### Vascular Evaluation

- Surveys the three vascular requirements of erection: **arterial inflow, engorgement and retention** → the two classic diagnoses, **arterial impairment** and **veno-occlusive dysfunction**.
- Counsel on purpose, alternatives, risks and benefits before any haemodynamic test.

### Combined Intracavernosal Injection and Stimulation

- The combined intracavernosal injection and stimulation (CIS) test is the **first-line evaluation of penile blood flow**: ICI of a vasodilator plus genital or audiovisual stimulation, with the response rated by an **independent assessor**.
- It **bypasses neural and hormonal influences**, testing penile vascular status directly.
- Regimens: **alprostadil alone 10–20 µg** (Caverject or Edex); **Bimix** (papaverine + phentolamine) 0.3 mL; **Trimix** (all three) 0.3 mL.

- **Technique:** 27–29-gauge needle at the lateral penile base into the corpus cavernosum → manual compression 5 minutes (prevents haematoma) → periodic rating of rigidity and duration → redose if the response is poor → flaccidity required before discharge → if no spontaneous detumescence within about 1 hour, intracavernosal diluted phenylephrine (500 µg/mL) every 3–5 minutes until flaccid.
- A sustained rigid erection = normal erectile haemodynamics → consider psychogenic, neurogenic or endocrine ED.
- **False positives** (normal-looking response despite disease) occur in up to a fifth of men with **borderline arterial inflow** (peak cavernous systolic flow 25–35 cm/s on duplex). **False negatives:** anxiety, needle phobia, inadequate dose.

### Duplex Ultrasonography (Gray-Scale or Color-Coded)

- Duplex after pharmacostimulation or CIS is **second-line** evaluation but the **most reliable and least invasive diagnostic modality** for ED; it adds imaging and quantification to the assessor's judgement.
- **High-resolution 7.5–12 MHz** real-time ultrasound with colour pulsed Doppler; dorsal and cavernous arteries, scanning from the **crura in the perineum to the tip**.
- Colour convention: **red towards the probe, blue away**.
- Protocol: **baseline velocities** → **injection** → **measurements every 5 minutes up to 20 minutes**; cavernous artery diameters; note communications between cavernous arteries or dorsal and cavernous arteries; rate erection quality simultaneously — a poor erection (often anxiety) prompts **redosing**, as for CIS.

### COMPARISON: DOPPLER WAVEFORM ACROSS THE PHASES OF ERECTION (FIGURE 72.6)

| Phase                         | Haemodynamics                                                      | Waveform                                                                                |
|-------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Flaccid                       | High resistance                                                    | Small systolic peaks                                                                    |
| Latent/filling (within 5 min) | <b>Low sinusoidal resistance</b>                                   | Waveform enlarges: <b>high forward flow in systole and diastole</b>                     |
| Tumescent                     | Intracavernous pressure rising                                     | <b>Diastolic velocity falls</b>                                                         |
| Full                          | High intracavernous pressure                                       | Sharp systolic peaks, slightly lower than in tumescence; forceful concentric pulsations |
| <b>Rigid</b>                  | <b>Intracavernous pressure exceeds systemic diastolic pressure</b> | <b>Diastolic flow zero or reversed</b> ; colour shifts red → blue                       |
| Detumescent                   | Pressure falls                                                     | Return towards baseline                                                                 |

*Artist's conception after intracavernous prostaglandin E<sub>1</sub> in a potent young male.*

- **Normative data:** in nonarteriogenic (psychogenic, neurogenic) ED, **peak systolic velocity (PSV) consistently exceeds 25 cm/s within 5 minutes**; mean PSV after pharmacostimulation in normal men **35–47 cm/s**; a **25 cm/s cut-off** is highly sensitive and specific against abnormal pudendal arteriography.
- In severe vascular ED, cavernous artery diameter increases **by less than 75%** and rarely exceeds 0.7 mm; but **percentage dilation does not correlate with arteriography, whereas PSV does**.
- **Anatomical variants confound interpretation:** early cavernous branching, multiple branches, distal perforators from dorsal or spongiosal arteries, and **site of measurement** (crus, proximal, mid-cavernosal) → risk of a false diagnosis of arteriogenic ED.
- **Asymmetry is significant:** a side-to-side PSV difference **>10 cm/s**, or **reversal of flow across a collateral**, suggests a significant atherosclerotic lesion.

### COMPARISON: DUPLEX CRITERIA — ARTERIAL INSUFFICIENCY VERSUS VENO-OCCLUSIVE DYSFUNCTION

| Parameter        | Cavernous arterial insufficiency | Veno-occlusive dysfunction                            |
|------------------|----------------------------------|-------------------------------------------------------|
| Defining problem | <b>Inadequate inflow</b>         | <b>Failure of maintenance despite adequate inflow</b> |

| Parameter                    | Cavernous arterial insufficiency                              | Veno-occlusive dysfunction                |
|------------------------------|---------------------------------------------------------------|-------------------------------------------|
| PSV                          | <25 cm/s (25–35 borderline; >35 normal)                       | Persistently >25 cm/s                     |
| End-diastolic velocity (EDV) | —                                                             | >5 cm/s                                   |
| Other                        | Systolic rise (acceleration) time >122 ms; asymmetry >10 cm/s | Rapid detumescence; resistive index <0.75 |
| Timing                       | Within 5 minutes                                              | 15–20 minutes after stimulation           |

- **Resistive index (RI) = (PSV – EDV) / PSV.** As intracavernous pressure reaches diastolic pressure, diastolic flow → 0 and RI → 1: RI >0.9 = normal; RI <0.75 = veno-occlusive dysfunction.

#### COMPARISON: TECHNICAL MODIFICATIONS OF PENILE SONOGRAPHY

| Modification                                                | Feature                                                                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Portable Midus-pulsed Doppler with laptop                   | In-office; records cavernous waveform reliably <b>without a real-time image</b>                             |
| Power Doppler                                               | Visualises distal cavernous ramifications <b>to arteriolar level</b>                                        |
| <b>Cavernous artery systolic occlusion pressure (CASOP)</b> | Doppler during <b>saline intracavernosal infusion</b> ; more invasive; tests integrity of cavernosal inflow |
| Oral PDE5I + visual erotic stimulation                      | <b>Noninvasive</b> stimulus alternative                                                                     |
| Postocclusive cavernous vasodilation                        | Reflects <b>endothelial function</b> ; diagnostic for organic ED                                            |
| Cavernous intima-media thickness                            | Possibly <b>more accurate than PSV</b> for vasculogenic ED                                                  |

Figure 72.5

- Colour duplex, longitudinal view: **collateral circulation** linking the right dorsal artery to the right and left cavernous arteries — the kind of communication that must be noted before interpreting velocities.

#### Dynamic Infusion Cavernosometry and Cavernosography

- Functional haemodynamic and radiographic assessment of the corpora; **third-line** evaluation.
- **Indications:** suspected **site-specific vasculogenic leak after perineal or pelvic trauma**, or **lifelong (primary) ED**; it generally **precedes corrective penile vascular surgery**.
- Technique: **two needles → intracavernosal pharmacological injection → simultaneous saline infusion and intracavernous pressure monitoring → maintenance flow rate, pressure decay and CASOP (confirming complete smooth muscle relaxation) → cavernosography to localise the leak.**
- **Veno-occlusive dysfunction = failure to raise intracavernous pressure to mean systolic blood pressure with infusion, or rapid pressure fall after infusion stops.**
- Cavernosography, normal: **corpora opacify with minimal or no venous or spongiosal filling**. Abnormal leakage sites: **glans, corpus spongiosum, superficial dorsal veins, cavernous and crural veins**.

Figures 72.7 and 72.8

- Figure 72.7 (third phase of DICC; four simultaneous traces: cavernosal flow by continuous-wave Doppler, brachial pressure, intracavernosal pressure, heparinised saline inflow): **CASOP is the intracavernosal pressure at which cavernosal pulsations return**. Brachial systolic 150 mm Hg versus CASOP 108 mm Hg → **gradient 42 mm Hg = abnormal**.
- Figure 72.8 (pharmacological cavernosography): a year after **penile fracture**, a cavernosum–spongiosum communication; in a 27-year-old with **primary impotence, crural venous leakage**.

#### Penile Angiography

- Anatomical study of the penile arterial tree; **third-line**. Reserved for the **young patient with traumatic arterial disruption** or **penile compression injury** being considered for **revascularisation**.

- Technique: **ICI of a vasodilator for maximal arterial dilatation → selective cannulation of the internal pudendal artery with contrast → iliac, internal pudendal and penile arteries documented; the inferior epigastric arteries** are studied to confirm suitability as donors.
- Marked intrapenile anatomical variation: the angiographer must separate **congenital variants from acquired lesions**.
- Figure 72.9: after pelvic injury (intracavernous papaverine 60 mg), patent common penile, dorsal and cavernous arteries on one side versus **nonvisualisation of the common penile artery** and branches on the other.

#### COMPARISON: THE VASCULAR TESTING LADDER

| Test               | Line   | Nature                             | Answers                                                     | Key indication                                |
|--------------------|--------|------------------------------------|-------------------------------------------------------------|-----------------------------------------------|
| <b>CIS</b>         | First  | Functional, office, assessor-rated | Can the vessels produce a rigid erection?                   | Screening vascular integrity                  |
| <b>Duplex</b>      | Second | Functional + imaging, quantified   | <b>Arterial versus veno-occlusive; asymmetry</b>            | <b>Most reliable, least invasive</b>          |
| <b>DICC</b>        | Third  | Invasive functional + radiographic | Is there a venous leak, and <b>where?</b>                   | Trauma, lifelong ED, before venous surgery    |
| <b>Angiography</b> | Third  | Invasive anatomical                | <b>Where is the arterial lesion; is the donor adequate?</b> | Young trauma patient before revascularisation |

#### Historical and Investigational Studies of Penile Blood Flow

#### COMPARISON: HISTORICAL AND INVESTIGATIONAL BLOOD FLOW STUDIES

| Test                                            | Principle                                                                                                                                             | Verdict                                                                                                |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Penile brachial pressure index (PBI)</b>     | Penile systolic ÷ brachial systolic pressure; paediatric cuff at base of <b>flaccid</b> penis, continuous-wave Doppler                                | <b>Invalid</b> — does not assess an induced erection; <b>not recommended</b>                           |
| Penile plethysmography (pulse volume recording) | 2.5–3-cm cuff to air plethysmograph; inflate above brachial systolic → deflate in 10-mm Hg steps recording waveforms; aggregate of all penile vessels | Performed on the <b>flaccid</b> penis — relevance questioned                                           |
| Radioisotopic penography                        | Penile blood volume after ICI using <b>technetium-99m-labelled red cells</b>                                                                          | <b>Poor correlation</b> with colour duplex                                                             |
| Penile MRI                                      | Anatomy and microcirculation; with angiography for iliac/penile vessels                                                                               | <b>Magnetic resonance angiography correlates well</b> with duplex                                      |
| Near infrared spectrophotometry                 | Continuous quantitative penile blood flow, with erectile stimulus                                                                                     | Documents haemodynamics; investigational                                                               |
| Cavernous smooth muscle content                 | Biopsy with light microscopy and computed morphometry                                                                                                 | Adjunct: <b>less smooth muscle, more collagen</b> in older men with veno-occlusive and arteriogenic ED |

**Concept** — *The vascular ladder climbs from function to anatomy and from office to angiography suite, and each invasive rung is justified only by a **surgical question**; any test performed on the **flaccid** penis measures the wrong physiological state.*

#### Psychophysiologic Evaluation

- Directly measures **tumescence and rigidity**; historically used to **separate psychogenic from organic ED** — a documented full erection proves an **intact neurovascular axis** and raises suspicion of a psychogenic cause.
- **Not first-line** because of technical and cost limitations; counsel on use, risks and benefits.

### Penile Tumescence and Rigidity Monitoring

- **Nocturnal penile tumescence (NPT)** monitoring studies sleep-related erections as a measure of physiological erectile ability.
- **Sleep-laboratory NPTR (the standard)**: number, tumescence (strain-gauge circumference change), maximal rigidity and duration of episodes, with **electroencephalography, electro-oculography and electromyography (EMG), nasal airflow and oxygen saturation** → documents **rapid eye movement (REM) sleep** (the period of true erectile events) and **hypoxia (sleep apnoea)**. **Axial rigidity** is measured on waking at maximal tumescence with a **buckling device at the tip**, plus photography.
- **RigiScan**: automated, portable; **two loops — penile base and coronal sulcus** — record tumescence (circumference) and **radial rigidity** by timed constrictions; office initialisation → home calibration. Normal: **four to five normal erectile episodes per night**. May predict veno-occlusive dysfunction.
- RigiScan limits: **radial rigidity does not predict axial rigidity; REM sleep cannot be verified**; wide variability even in normal men.
- **NPT electrobioimpedance (NEVA)**: three electrodes (hip, penile base, glans) and a thigh recorder; an imperceptible current glans → hip ground; **impedance falls with tumescence**, measuring volume and length change. No REM monitoring; wearable home sensors emerging but the **volume-rigidity relationship** is unproven.

### COMPARISON: NOCTURNAL ERECTION MONITORING METHODS

|               | Sleep-laboratory NPTR            | RigiScan               | NEVA                          |
|---------------|----------------------------------|------------------------|-------------------------------|
| Setting       | Laboratory                       | Home                   | Home                          |
| Rigidity      | <b>Axial</b> (buckling)          | <b>Radial</b>          | None — <b>volumetric</b> only |
| Sleep staging | <b>REM and apnoea documented</b> | No                     | No                            |
| Sensor        | Strain gauges + polysomnography  | Two constricting loops | Three impedance electrodes    |

Figure 72.10

- RigiScan: (A) at least two **well-sustained, completely rigid** nocturnal erections — normal; (B) two **poorly sustained, poorly rigid** episodes. **Home studies do not document sleep quality**.
- **Summary**: NPTR objectively evaluates the somatic basis of erection, theoretically **free of psychological interference**, but it **does not give cause or severity**, is poorly reproducible, and may not reflect **wakeful, sexually relevant erections**.
- **False positives** in conditions such as **multiple sclerosis**; **false negatives** in **ageing, depression and anxiety** (which alter sleep-related erections).
- Reserve for special circumstances: **obscure cause where noninvasive testing is desired**.

### Audiovisual and Vibratory Stimulation

- Erotic video stimulation with monitoring is a **reliable, time- and cost-effective alternative to NPTR** for separating organic from psychogenic ED, and **more physiological** (consistent with waking erectile behaviour).
- **False negatives** with **endocrine abnormalities**; **false positives** with psychological **inhibition of erotic excitement**.
- Combinable with pharmacological erection testing and RigiScan.

### COMPARISON: NPTR VERSUS AUDIOVISUAL STIMULATION

|              | NPTR                                            | AVSS                                                  |
|--------------|-------------------------------------------------|-------------------------------------------------------|
| State tested | <b>Sleep</b> — psychological inhibition removed | <b>Awake</b> — inhibition bypassed by erotic stimulus |

|                         | NPTR                                                                            | AVSS                                                                     |
|-------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Physiological relevance | Questioned (not sexually relevant)                                              | <b>More physiological</b>                                                |
| Cost and time           | Higher                                                                          | <b>Lower</b>                                                             |
| Misleading results      | False negative: ageing, depression, anxiety; false positive: multiple sclerosis | False negative: endocrine disease; false positive: excitement inhibition |

### Neuroimaging

- **PET and functional MRI**, with video stimulation or an erectogenic drug (oral apomorphine), map the central arousal network: **anterior cingulate, insula, amygdala, hypothalamus and secondary somatosensory cortices**.
- **Functional brain abnormalities in psychogenic ED** suggest a **biological basis** for the diagnosis. Clinical role not yet defined.

**Concept** — Every psychophysiologic test asks one question: can this neurovascular axis produce a rigid erection when psychological inhibition is **removed (sleep)** or **bypassed (erotic stimulus)**? A yes points to psychogenic ED; a no cannot say why.

### Psychological Evaluation

- Addresses psychological and interpersonal factors; population studies link ED with **anxiety, depression, low self-esteem, negative outlook, emotional stress and sexual coercion history**.
- The urologist can initiate it simply; the **Whooley two-question depression screen**: during the past month, have you often been bothered by **feeling down, depressed or hopeless?** by **little interest or pleasure in doing things?**
- **The diagnostic interview is central**, handled straightforwardly.

### COMPARISON: PSYCHOLOGICAL CAUSES OF SEXUAL DYSFUNCTION

| Readily discernible                                                                                                                                                                                                       | Less immediately discernible                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Fear of failure; <b>performance anxiety</b> (in widowers: dating, new partners, unresolved mourning/guilt); insufficient stimulation; loss of attraction; adjustment to chronic illness or surgery; relationship conflict | Unresolved parental attachments; sexual identity issues; <b>sexual trauma</b> ; extramarital affairs; cultural–religious taboos |

- **Primary psychogenic ED**: suspect in the **absence of organic risk factors**; supported by **noncoital erections** (masturbatory, nocturnal, on waking).
- Subtypes: **generalised versus situational; lifelong (primary) versus acquired (secondary)** — including substance abuse or major psychiatric illness).
- **Relationship factors** may cause psychogenic ED or exacerbate organic ED: intimacy and trust, status and dominance, loss of attraction, ability to achieve satisfaction without erection, communication. Interview the patient alone, the couple together and partners separately.
- **Complex intrapsychic causes** (trauma, cultural or religious strife, compulsive sexual behaviour, neurosis) and psychiatric comorbidity (substance abuse, depression, anxiety, personality disorder) → **refer to a psychological expert**.

### Neurologic Evaluation

- Targets **peripheral, spinal and supraspinal** centres and **somatic and autonomic** pathways governing erection.
- Tests have had **limited impact on routine management** — mostly for **research and medicolegal work** — and suffer poor sensitivity, reproducibility, reliability and validity, **especially autonomic tests**; somatic testing is reproducible and valid. Tests of **neurotransmitter release**, the most relevant, are undeveloped.

**COMPARISON: NEUROPHYSIOLOGICAL TESTS FOR ERECTILE DYSFUNCTION**

| Test                                                                       | Pathway                                       | Method                                                                                                                                             | Finding / verdict                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Biothesiometry</b>                                                      | Somatic afferent                              | Handheld electromagnetic vibrator on index finger pulps, both sides of shaft, glans;<br><b>vibratory perception threshold</b>                      | Does not accurately portray dorsal penile nerve function (glans skin responds poorly to vibration)                                                                                                                                |
| <b>Bulbocavernosus reflex latency</b> (sacral evoked response)             | Somatic reflex arc                            | Square-wave stimuli via <b>penile ring electrodes; concentric needle electrodes in both bulbocavernosus muscles</b> ; stimulus-to-response latency | Abnormal latency → probable neuropathology; but a <b>full limb-nerve battery is more sensitive</b> than pudendal-specific tests                                                                                                   |
| <b>Dorsal nerve conduction velocity</b>                                    | Somatic                                       | Stimulating electrodes at glans and base → two reflex latencies                                                                                    | Abnormal velocity diagnostic of neurogenic ED <b>in diabetes</b>                                                                                                                                                                  |
| Genitocerebral evoked potential                                            | Somatic afferent, spinal and supraspinal      | Dorsal nerve stimulation; recording over sacral cord and cortex                                                                                    | Poor latency discrimination; useful for subtle afferent deficits                                                                                                                                                                  |
| Heart rate variability; <b>sympathetic skin response</b>                   | Autonomic (parasympathetic; sympathetic)      | Heart rate in quiet/deep breathing and leg raising; shock at median or tibial nerve, recording at hand, foot or <b>penis</b>                       | Penile recording potentially useful for autonomic innervation                                                                                                                                                                     |
| Penile thermal sensory testing                                             | Small sensory (autonomically affected) fibres | Thermal threshold                                                                                                                                  | Correlates well with clinical neurogenic ED                                                                                                                                                                                       |
| Electrochemical skin conductance (Sudocan)                                 | Sympathetic (sudomotor)                       | Skin conductance                                                                                                                                   | Promising in DM; <b>more sensitive than sympathetic skin response</b> ; needs trials                                                                                                                                              |
| <b>CC-EMG</b> / single potential analysis of cavernous electrical activity | Autonomic (cavernous)                         | Direct cavernous recording                                                                                                                         | Normal: rhythmic slow waves with bursts when flaccid, <b>silence during tumescence</b> ; autonomic neuropathy → <b>discordant persistent activity</b> .<br>Normal peak-to-peak 120–500 µV, mean duration 12 s; utility questioned |

- **Diabetic penile neuropathy can exist without distal (foot) neuropathy**, so examining the feet cannot exclude it.

**Concept** — Neurophysiology splits along one line: **somatic tests are reproducible and valid, autonomic tests are not** — yet erection is an autonomic event, so the tests that work best measure the pathway that matters least.

### Hormonal Evaluation

- Endocrinopathies relevant to erection: **T deficiency** (formerly hypogonadism), **hyperthyroidism** and **DM** (altered androgen modulation).
- Screening questionnaires and the validated hypogonadism screener **lack specificity (and sometimes sensitivity)**; the central feature is **biochemical testing**.

### Serum Testosterone Measurements

- Low T is found in a **variable minority** of men presenting with ED, depending on the population.

- Age-related terms: androgen deficiency of the aging male (ADAM), partial ADAM (PADAM), hypoandrogenism, symptomatic late-onset hypogonadism (SLOH), andropause. **Adult-onset hypogonadism (AOH)** ties T deficiency to **adult-onset health conditions rather than ageing per se**.
- **Circulating fractions:** free 0.5–3%; tightly bound to sex hormone-binding globulin (SHBG) ≈30%; loosely bound to albumin ≈67%. Bioavailable T = free + albumin-bound.

#### COMPARISON: CONDITIONS ALTERING SHBG

| SHBG decreased → bioavailable T raised                                                                                 | SHBG increased → bioavailable T lowered                                                                |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Moderate <b>obesity</b> ; nephrotic syndrome; <b>hypothyroidism</b> ; glucocorticoids, progestins, androgenic steroids | <b>Ageing</b> ; hepatic cirrhosis; <b>hyperthyroidism</b> ; HIV infection; anticonvulsants; oestrogens |

- Low SHBG accompanies insulin resistance, but SHBG is **variable in DM** (obesity, age) → **diagnose hypogonadism in DM by low bioavailable T**.
- Free T is theoretically most relevant, but **commercial free-T assays are unreliable**; the **best indicator is calculated bioavailable T** — useful for dose adjustment when total T is low-normal. The International Society for the Study of the Aging Male (ISSAM) formula needs **total T and SHBG**, plus **albumin** in serious liver disease or hypoalbuminaemia.
- **Screening: total T is sufficient**, drawn **7:00–11:00 AM** (diurnal peak in young and middle-aged men). **Reference range 280–1000 ng/dL**; the treatment threshold remains unresolved.
- Only a **low threshold of T is needed for sexual function**: correlation is weak above about **8 nmol/L (≈230 ng/dL)**; the AUA deficiency threshold is **300 ng/dL (≈10.4 nmol/L)**.
- Low or low-normal T → **repeat to confirm**; about a third of mildly abnormal values are **normal on repeat**. Carrier-protein conditions prompt further testing.
- **Oestradiol (E<sub>2</sub>)**: role in libido, spermatogenesis and erection; higher in ED; measure when there are **breast symptoms or gynaecomastia**.

#### Serum Gonadotropin Measurements

- With the **second total T**, add **LH and prolactin**; gonadotropins **localise** the lesion in the hypothalamic–pituitary–gonadal axis.

#### COMPARISON: PRIMARY VERSUS SECONDARY HYPOGONADISM

|                                     | Primary (testicular failure)                          | Secondary (central)                           |
|-------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| Testosterone                        | Low                                                   | Low                                           |
| LH and follicle-stimulating hormone | <b>Raised</b> — appropriate response to lost feedback | <b>Normal or low</b> — inappropriate response |
| Next step                           | Testicular cause                                      | <b>Prolactin; pituitary MRI</b> if indicated  |

#### Serum Prolactin Measurement

- Hyperprolactinaemia suppresses **gonadotropin-releasing hormone** → loss of **pulsatile LH** → low T; independently it may impair **peripheral T → dihydrotestosterone (DHT) conversion**, causing loss of libido regardless of T.
- Suspect with **low T and low or inappropriately normal LH**. Routine prolactin in ED is **controversial** (low yield versus missed cases).
- Causes: **antipsychotics, tricyclics, opiates**; prolactin-secreting tumours; **hypothyroidism**; hypothalamic lesions; renal insufficiency; cirrhosis; chest wall lesions.

#### Magnetic Resonance Imaging Scans

- Central (hypogonadotropic) hypogonadism and unexplained hyperprolactinaemia → **pituitary MRI**.

- Indications: **severe central hypogonadism (T <150 ng/dL)**; suspected pituitary disease — **panhypopituitarism, persistent hyperprolactinaemia, tumour mass effect.**

### Serum Thyroid Function Tests

- **Hyperthyroidism** causes ED via increased **aromatisation of T to oestrogen** (raising SHBG) or increased **adrenergic tone** (smooth muscle contraction, psychobehavioural effects).
- Symptoms: hyperactivity, irritability, heat intolerance, palpitations, fatigue, weight loss; signs: tachycardia, tremor, goitre, lid retraction.
- Diagnosis: **high total or free thyroxine (T<sub>4</sub>) or triiodothyronine (T<sub>3</sub>) with low thyroid-stimulating hormone (TSH).**

#### KEY POINTS: DIAGNOSTIC EVALUATION

- Basic evaluation: **detailed history, physical examination, proper laboratory tests.**
- The sexual history defines onset, duration, conditions, severity and cause.
- **Cardiac risk assessment and risk reduction** for all men undergoing ED evaluation.
- Hormonal evaluation: T deficiency, hyperthyroidism and DM.
- Questionnaires document presence, severity and treatment response.
- Specialised testing for **complicated or atypical presentations.**

**Concept** — Hormonal work-up is a **localisation exercise**: total T screens, calculated bioavailable T corrects for carrier-protein distortion, LH and prolactin place the lesion in testis or pituitary, and MRI is reserved for the central lesion that could be a tumour.

## TREATMENT CONSIDERATIONS

- Treatment follows appropriate work-up; interventions are **etiologically specific or nonspecific**, and a **cause-specific intervention ideally offers correction.**
- **Shared decision making**: all options presented with advantages and disadvantages, risks and benefits weighed for the individual (Figure 72.1).

### Lifestyle Modifications

- ED risk tracks **preventable or treatable comorbidities** (DM, CVD, metabolic syndrome); optimising them may prevent or ameliorate ED.
- **Smoking cessation** is associated with recovery of erectile function.
- **Increased exercise** benefits sedentary men with ED.
- **Randomised trial**: obese men with moderate ED and no overt CVD improved IIEF scores with **exercise and weight control** versus education alone, with falls in BMI and C-reactive protein and rise in activity.
- **Mediterranean-style diet and caloric reduction** improve erectile function in metabolic syndrome.
- **No-nose bicycle saddle** restored erectile function in occupational cyclists (perineal trauma relieved).
- Supported by systematic reviews and meta-analyses; mechanisms: **reduced CV risk factors, raised serum T, improved mood and self-esteem.**

### Medication Change

- Once a drug is implicated, change dose or class — this may reverse ED.
- Antihypertensives: switch **thiazides and  $\beta$ -blockers** → **calcium channel blockers or renin-angiotensin system inhibitors** (angiotensin-converting enzyme inhibitors, angiotensin receptor blockers).

- **SSRI sexual dysfunction** (ED, delayed ejaculation): **substitution (bupropion, nefazodone, buspirone, mirtazapine), drug holidays, dose reduction, watchful waiting, or add a PDE5I.**

## Psychosexual Therapy

- Justified by the interplay of psychological and interpersonal factors, but **well-controlled outcome evidence is lacking**; in practice an ill-defined mix of behavioural, relational, psychoanalytic and cognitive approaches.
- Interventions: **systematic desensitisation, sensate focus**, interpersonal therapy, cognitive-behavioural therapy, sex education, couples' communication and sexual skills training, masturbation exercises.
- **Integrated treatment** (psychosexual therapy + oral therapy, ICI or vacuum device) succeeds, especially with **motivational obstacles**.
- For mild-to-moderate issues the urologist can use a **biopsychosocial model** (awareness plus counselling on normal function and acceptable behaviour); intensive therapy needs a mental health clinician.

## Hormonal Therapy

- Prescribed when a hormonal disturbance is identified. The **urologist treats T deficiency and hyperprolactinaemia**; endocrinologists lead for other endocrinopathies.

## Testosterone Therapy

- Goal: **normal 24-hour levels resembling the diurnal pattern**. Measure T before and during treatment, but **judge efficacy by clinical response**, not a precise level.
- **Short (≈3-month) therapeutic trial; stop if no response.**
- Adverse effects: **erythrocytosis**, sleep apnoea, urinary symptoms, prostate cancer progression risk, gynaecomastia, acne.
- CV safety: a meta-analysis of all placebo-controlled randomised controlled trials (RCTs) **did not support a causal relationship between T and CV events**.
- Monitoring sequence: **baseline digital rectal examination (DRE) and PSA, haemoglobin/haematocrit, liver function tests, cholesterol and lipid profile → efficacy and AE review at 3–6 months → annually.**
- Prefer **short-acting preparations initially**, so therapy can be stopped promptly after an AE.

**TABLE 72.6 TESTOSTERONE PREPARATIONS**

*Modified from Corona et al., 2011; doses from the table and text combined*

| Route / agent                                   | Chemistry;<br>half-life     | Dose                                                                                                            | Advantages                                                        | Disadvantages                                                             |
|-------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Oral</b>                                     |                             |                                                                                                                 |                                                                   |                                                                           |
| T undecanoate<br>(Jatenzo, Tlando,<br>Kyzatrex) | 17β-hydroxyl<br>ester; 4 h  | Jatenzo 158–396 mg twice<br>daily; Tlando 225 mg twice<br>daily; Kyzatrex 100 mg daily to<br>400 mg twice daily | Oral; adjustable;<br><b>partly escapes<br/>hepatic first pass</b> | <b>Raised blood pressure</b>                                              |
| Methyltestosterone                              | 17α-<br>alkylated;<br>3.5 h | 10–50 mg 2–3 times daily                                                                                        | Oral; adjustable                                                  | <b>Hepatotoxic; obsolete</b>                                              |
| Mesterolone                                     | 1α-alkylated;<br>8 h        | 25–50 mg 2–3 times daily                                                                                        | Oral; adjustable                                                  | <b>Not aromatisable to<br/>oestrogen</b>                                  |
| <b>Intramuscular</b>                            |                             |                                                                                                                 |                                                                   |                                                                           |
| T enanthate                                     | 17β-ester;<br>4–5 days      | 200–250 mg every 2–3 weeks                                                                                      | Cheap; adjustable                                                 | <b>Wide fluctuations;<br/>repeated injections;<br/>more polycythaemia</b> |

| Route / agent                       | Chemistry;<br>half-life           | Dose                                                                                                           | Advantages                                           | Disadvantages                                              |
|-------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------|
| T cypionate                         | 17 $\beta$ -ester; 8 days         | 200 mg every 2–3 weeks                                                                                         | Cheap; adjustable                                    | Wide fluctuations; repeated injections                     |
| T propionate                        | 17 $\beta$ -ester; <b>20 h</b>    | Table: 100 mg every 2 days<br>Text: 200 mg every 2–3 days                                                      | Cheap                                                | Wide fluctuations; frequent injections; more polycythaemia |
| T undecanoate depot (Aveed; Nebido) | 17 $\beta$ -ester; <b>34 days</b> | <b>750 mg at start, 4 weeks, then every 10 weeks (US);</b> 1000 mg every 10–14 weeks (Europe; not US-approved) | Levels within normal range; <b>long-lasting</b>      | Injection-site pain                                        |
| <b>Subcutaneous</b>                 |                                   |                                                                                                                |                                                      |                                                            |
| Pellets (Testopel, 75 mg each)      | Native T                          | 2–6 pellets (150–450 mg) every 3–6 months                                                                      | 3–4 treatments/year                                  | Invasive placement; <b>extrusion, site infection</b>       |
| <b>Buccal</b>                       |                                   |                                                                                                                |                                                      |                                                            |
| Striant                             | Native T; 12 h                    | 30 mg twice daily; mucoadhesive, to gum above incisors                                                         | Physiological levels                                 | Oral irritation; twice-daily dosing; unpleasant taste      |
| <b>Transdermal and intranasal</b>   |                                   |                                                                                                                |                                                      |                                                            |
| Patches (Androderm; Testoderm TTS)  | Native T; 10 h                    | 2.5–10 mg/day                                                                                                  | <b>Mimic circadian rhythm</b> ; simple               | <b>Skin irritation</b> ; daily                             |
| Gel 1–2% (AndroGel, Testim)         | Native T; 6 h                     | 40.5–100 mg/day                                                                                                | Normal range; flexible; less irritation than patches | <b>Transfer by intimate contact</b> ; daily                |
| Solution 2% (Axiron)                | Native T                          | 60–120 mg/day (30 mg per axilla)                                                                               | Normal range                                         | Transfer risk; daily                                       |
| Nasal gel (Natesto)                 | Native T; <b>1 h</b>              | 11 mg (one actuation per nostril) 2–3 times daily (22–33 mg/day)                                               | Quick onset; <b>no transference</b>                  | Nasal irritation; 2–3 doses daily                          |

- **What the table is for:** choosing a route by pharmacokinetic profile. Short half-life → frequent dosing but easy withdrawal; long half-life → convenience but slow reversal.
- **Pattern:** injectable esters fluctuate (supraphysiological peaks, low troughs, polycythaemia); transdermal agents best mimic the circadian rhythm; **alkylated oral agents are hepatotoxic**, while oral undecanoate is safe but raises blood pressure.
- **Trap:** nasal T is the only topical preparation with **no transfer risk**; gels and solutions carry transfer risk to partners and children.

#### Intramuscular

- Enanthate/cypionate 200–250 mg every 2–3 weeks → **supraphysiological for 72 hours, declining to subphysiological by 10–12 days**; for trough-related mood or sexual swings, **100 mg every 7–10 days**.

#### Transdermal

- Morning application → higher initial absorption → **mimics diurnal variation**.
- **Testoderm scrotal patch** (4–6 mg, no adhesive) failed: application difficulty, scrotal shaving, and **high DHT from abundant scrotal 5 $\alpha$ -reductase**. Testoderm TTS (5 mg; arm, back, buttock) and Androderm (2.5 or 5 mg) followed: itching, chronic irritation, contact dermatitis → **topical corticosteroid cream, rotate sites, avoid sun-exposed skin**.

- **AndroGel 1%** (50, 75, 100 mg) — only about **10% absorbed over 24 h**; Testim 1% (5-g tube, 50 mg). Apply each morning to clean dry skin of shoulders, upper arms or abdomen; allow to dry before dressing.
- **Axiron 2% solution**: 30 mg to each axilla daily by metered applicator — concealable, **highly permeable**, **high 5 $\alpha$ -reductase**.

### Oral

- Historically limited by **hepatotoxicity of alkylated forms** (hepatitis, cholestatic jaundice, hepatomas, haemorrhagic liver cysts, hepatocellular carcinoma) and the **large doses (>200 mg/day)** needed because of **first-pass hepatic inactivation**.
- **17 $\alpha$ -methyltestosterone and fluoxymesterone should not be prescribed**.
- **T undecanoate** (T esterified with undecanoic acid) is safe because it **partly escapes hepatic inactivation**. The new US oral formulations (doses in Table 72.6) sustain eugonadal levels; **monitor blood pressure**.

### Alternative Hormone Treatments

#### COMPARISON: ALTERNATIVE HORMONAL THERAPIES

| Agent                                  | Rationale                                                                                                                    | Limitation                                                                                                                                             |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHT gel (125–250 mg/day)               | Pure androgen, <b>not aromatisable</b> → no oestrogenic effect on prostate growth or lipids; physiological-equivalent levels | Not recommended — insufficient evidence                                                                                                                |
| Dehydroepiandrosterone (DHEA)          | Androgen- and oestrogen-like supplement                                                                                      | Little evidence for sexual benefit; DHEA and other precursors (DHEA-S, androstenediol, androstenedione) may <b>stimulate breast or prostate cancer</b> |
| Human chorionic gonadotropin (hCG)     | Raises total and free T and E <sub>2</sub> ; improves body composition                                                       | <b>No documented sexual-function benefit</b>                                                                                                           |
| Antioestrogens; aromatase inhibitors   | Raise endogenous T                                                                                                           | Insufficient evidence                                                                                                                                  |
| Selective androgen receptor modulators | Under development                                                                                                            | —                                                                                                                                                      |

*None is currently recommended for older men with T deficiency.*

### Hyperprolactinemia Treatments

- **T replacement is neither corrective nor sufficient; treat the cause**.
- Stop offending drugs (**oestrogens, morphine, sedatives, neuroleptics**).
- Prolactinoma: medical first, surgical if needed. **Bromocriptine** (dopamine agonist) lowers prolactin, restores T and **shrinks the tumour**; **neurosurgical ablation** if medication fails or **visual effects from optic nerve compression**.
- Erectile recovery is most evident when prolactin was **markedly raised (>40 ng/mL)**.

**Concept** — *Hormonal therapy works only when the hormone is the cause: testosterone restores erection in true deficiency, is futile in hyperprolactinaemia where the **prolactin must be corrected**, and precursor or alternative hormones carry risk without proven sexual benefit.*

### Pharmacologic Therapies

- Pharmacotherapies mimic the natural biochemistry of erection: they **promote proerectile** or **oppose antierectile** mechanisms at peripheral and central levels.
- Regulatory approval standards now define which commercially developed therapies qualify.

**COMPARISON: PHARMACOLOGICAL STRATEGIES BY LEVEL AND DIRECTION**

| Level                                          | Promote proerectile mechanisms                                                                                                                                                         | Oppose antierectile mechanisms                                           |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Peripheral</b><br>(corporeal smooth muscle) | Receptor agonists or effectors that raise <b>cyclic guanosine monophosphate (cGMP)</b> or <b>cyclic adenosine monophosphate (cAMP)</b> ; <b>inhibit PDEs</b> that terminate relaxation | <b><math>\alpha_1</math>-adrenergic antagonists</b> reducing contraction |
| <b>Central</b> (brain, spinal cord)            | <b>Dopamine D<sub>2</sub> agonists</b> (medial hypothalamus)                                                                                                                           | <b>5-HT<sub>1A/2</sub> (serotonergic) antagonists</b> (spinal cord)      |

**Oral Therapy**

- Meets the attributes of **ideal therapy** — **convenience, simplicity, noninvasiveness** — increasingly with efficacy.

*Phosphodiesterase Type 5 Inhibitors*

- US Food and Drug Administration (FDA) approvals: **sildenafil 1998** (Viagra); **varденаfil and тaдалaфil 2003** (Levitra, Cialis); **avanafil 2012** (Stendra).
- Mechanism: **sexual stimulation → nitric oxide (NO) from nerve endings and endothelium → cGMP → cavernosal smooth muscle relaxation**; PDE5 degrades cGMP, and PDE5Is block this.
- They **augment but do not induce** erection — **sexual stimulation is obligatory**. Selectivity arises from the **high PDE5 concentration in cavernosal smooth muscle**.
- Structure: guanine-like base, ribose-like system and phosphodiester-like bond fit the catalytic site. **Sildenafil and vardenafil are similar; тaдалaфil differs; avanafil differs most**.
- **PDE6 (retina)**: sildenafil and vardenafil cross-react more → **visual disturbance**. **PDE11**: cross-reacted by **тадалaфil** alone (significance unclear). Other side effects reflect PDE5 in vascular and gastrointestinal smooth muscle.
- **Tadalafil's long half-life** gives the widest therapeutic window.

**TABLE 72.7 COMPARISON OF FOUR PHOSPHODIESTERASE TYPE 5 INHIBITORS CURRENTLY AVAILABLE IN THE UNITED STATES**

|                                               | <b>Sildenafil</b>  | <b>Vardenafil</b>                                 | <b>Tadalafil</b> | <b>Avanafil</b>    |
|-----------------------------------------------|--------------------|---------------------------------------------------|------------------|--------------------|
| Peak concentration (C <sub>max</sub> , ng/mL) | 450                | 20.9                                              | 378              | 2153               |
| Time to peak (T <sub>max</sub> , h)           | 0.8                | 0.7–0.9                                           | <b>2</b>         | <b>0.3–0.5</b>     |
| Onset (min)                                   | 15–60              | 15–60                                             | 15–120           | 15–60              |
| Half-life (h)                                 | 3–5                | 4–5                                               | <b>17.5</b>      | 3–5                |
| Bioavailability                               | 40%                | 15%                                               | Not tested       | 30%                |
| Fatty food                                    | Reduced absorption | Reduced absorption                                | <b>No effect</b> | Reduced absorption |
| Doses (mg)                                    | 25, 50, 100        | 5, 10, 20                                         | 5, 10, 20        | 50, 100, 200       |
| Headache, dyspepsia, flushing                 | Yes                | Yes                                               | Yes              | Yes                |
| Backache, myalgia                             | Rare               | Rare                                              | <b>Yes</b>       | Rare               |
| Blurred/blue vision                           | <b>Yes</b>         | Table: rare · Text: more common (with sildenafil) | Rare             | <b>No</b>          |
| Antiarrhythmic precaution                     | No                 | <b>Yes</b>                                        | No               | No                 |
| Nitrate contraindication                      | Yes                | Yes                                               | Yes              | Yes                |

*Vision row: the table grades vardenafil "rare", while the text (randomised trials) groups vardenafil with sildenafil for visual effects via PDE6 cross-reactivity — the table reflects frequency, the text relative frequency against tadalafil and avanafil.*

- **Pattern:** tadalafil is the outlier on every axis — slowest peak, longest half-life, **food-independent**, back pain rather than visual effects. Avanafil is fastest to peak and free of visual effects.
- **Trap:** the one row identical across all four is **nitrate contraindication**; vardenafil alone carries the **antiarrhythmic/QT** precaution.
- **Efficacy is equivalent** across all four; **no superiority can be claimed** without head-to-head trials. Successful intercourse  $\approx 70\%$ ; lower (40–50%) in **DM** and after radical prostatectomy generally; approaching 60–70% after **bilateral nerve-sparing** prostatectomy.
- **On-demand dosing 30–60 minutes before activity**, matching time to peak:  $\approx 30$  min avanafil, 1 h sildenafil and vardenafil, 2 h tadalafil. Onset may be within 20 minutes, but patients value **hardness and maintenance** more than speed. **Daily tadalafil** is approved for convenience.
- **Optimisation:** adequate sexual stimulation (NO release), **reduced food intake**, **dose escalation**, **repeated attempts (up to 9–10 before declaring failure)**, and correcting glycaemia, lipids and androgen status.
- Patient and partner satisfaction well demonstrated; **long-term adherence is poor** (up to nearly half stop) — psychosocial factors and repeated dosing.
- T + PDE5I: **controversial** evidence of improved response. **Tadalafil 5 mg daily + sildenafil 50 mg** on demand for severe ED without extra side effects.

### BOX 72.1 WARNINGS AND DRUG INTERACTIONS

Package inserts of all four PDE5Is (September 2013)

| Category              | Content                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Warned against        | <b>Severe CVD; left ventricular outflow obstruction</b> (aortic stenosis, idiopathic subaortic stenosis); severely impaired autonomic blood pressure control                                                                                                                                                                                                                                                                                   |
| Not studied in trials | Myocardial infarction, stroke or life-threatening arrhythmia <b>within 6 months</b> ; <b>New York Heart Association class <math>\geq</math> II</b> heart failure or unstable angina; resting <b>hypotension <math>&lt;90/50</math></b> or <b>hypertension <math>&gt;170/100</math> mm Hg</b> ; <b>retinitis pigmentosa</b> and hereditary retinal degenerations; <b>Child-Pugh C</b> liver disease; <b>end-stage renal disease on dialysis</b> |
| CYP3A4 inhibitors     | <b>Ketoconazole, itraconazole, ritonavir</b> $\rightarrow$ raised PDE5I levels $\rightarrow$ <b>reduce dose</b>                                                                                                                                                                                                                                                                                                                                |
| CYP3A4 inducers       | <b>Rifampicin</b> $\rightarrow$ faster breakdown $\rightarrow$ <b>higher dose</b>                                                                                                                                                                                                                                                                                                                                                              |
| Organ dysfunction     | Renal or hepatic impairment may need dose adjustment                                                                                                                                                                                                                                                                                                                                                                                           |

- **CV safety is well demonstrated:** no excess myocardial infarction or death versus expected rates; no worsened ischaemia, coronary vasoconstriction or haemodynamics in known coronary disease or heart failure. Nonetheless, **CV risk assessment and stabilisation before starting**.
- Caution: **aortic stenosis, left ventricular outflow obstruction, hypotension, hypovolaemia**. Minimal QTc effect; **vardeafil** not recommended with **class 1A (quinidine, procainamide)** or **class 3 (sotalol, amiodarone)** antiarrhythmics or **congenital long QT**.
- **Nitrates in any form — absolute contraindication:** sublingual glyceryl trinitrate, isosorbide dinitrate, other antianginal nitrates, **amyl nitrite/nitrate "poppers"**. Nitrate use **more than 2 weeks earlier** is not a contraindication.

- **Angina during sex on a PDE5I:** stop → emergency care → tell staff a PDE5I was taken → **no nitroglycerin for 24 h (sildenafil, vardenafil) or 48 h (tadalafil)**. In acute myocardial infarction, all usual therapy **except organic nitrates**.
- **PDE5I-induced hypotension:** Trendelenburg position → intravenous fluids →  $\alpha$ -adrenergic agonist (phenylephrine) → intra-aortic balloon counterpulsation if refractory (American College of Cardiology/American Heart Association). **No pharmacological antidote** to the PDE5I–nitrate interaction.
- Caution with  $\alpha$ -adrenergic blockers — additive blood-pressure lowering.
- Side effects: **headache (commonest)**, dyspepsia, flushing, nasal congestion, myalgia/back pain, visual disturbance (photophobia, blue vision). **Flushing and visual effects more with sildenafil/vardenafil; back pain/myalgia more with tadalafil**. Mild, abate with time; few discontinue.
- **Nonarteritic anterior ischaemic optic neuropathy (NAION):** systematic reviews show **no increased risk**; reported cases had vascular risk factors. Still advise **stopping and seeking immediate care after sudden visual loss**.
- **Melanoma** association is confounded (surveillance bias); **prostate cancer recurrence risk refuted**.
- **Penile rehabilitation:** scheduled PDE5I to promote recovery of spontaneous erections, chiefly after radical prostatectomy — **role unclear** for lack of well-designed trials.

#### COMPARISON: TRIALS OF SCHEDULED PDE5I FOR ERECTILE RECOVERY

| Trial                                 | Setting and regimen                                                                    | Comparator | End point                                        | Verdict                         |
|---------------------------------------|----------------------------------------------------------------------------------------|------------|--------------------------------------------------|---------------------------------|
| Sildenafil nightly trial              | Radical prostatectomy; nightly sildenafil 36 weeks from week 4                         | Placebo    | Erections sufficient for sex at $\approx 1$ year | <b>Positive</b>                 |
| <b>REINVENT</b> (vardenafil)          | Radical prostatectomy; vardenafil nightly or on demand 9 months from day 14            | Placebo    | Recovery at $\approx 1$ year                     | <b>Negative</b> — no difference |
| Sildenafil nightly vs on-demand trial | Radical prostatectomy; sildenafil nightly versus on demand, 12 months, 1-month washout | On demand  | Recovery                                         | <b>Negative</b> — no difference |
| Sildenafil radiotherapy trial         | <b>Radiotherapy</b> ; daily sildenafil 6 months                                        | Placebo    | Mild or no ED at 12 months                       | <b>Positive</b>                 |
| Other ED contexts                     | Continuous PDE5I then withdrawal                                                       | —          | Sustained natural function                       | <b>Negative</b>                 |

- **Combining PDE5Is with vasoactive penile pharmacotherapy** is off-label; precautions advised.

#### Other Oral Agents

#### COMPARISON: NON-PDE5I ORAL AGENTS

| Agent                        | Target and site                                                                    | Dose and use                                         | Adverse effects                              | Status                                                                          |
|------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------|
| <b>Phentolamine mesylate</b> | Nonselective $\alpha_1/\alpha_2$ antagonist; peripheral (postsynaptic $\alpha_1$ ) | 40 mg; efficacy $\approx 40\%$ in minimal ED         | Headache, flushing, nasal congestion         | Adequacy for severe ED unproven                                                 |
| <b>Yohimbine</b> (Yocon)     | <b>Central <math>\alpha_2</math> antagonist;</b> yohimbe bark alkaloid             | <b>5.4 mg three times daily</b> for at least a month | Hypertension, anxiety, tachycardia, headache | Beats placebo overall but <b>not in organic ED</b> → <b>psychogenic ED only</b> |

| Agent                                           | Target and site                                                                                  | Dose and use                                                                            | Adverse effects                                                                                                                                            | Status                                       |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Apomorphine</b><br>(Uprima)                  | Central <b>D<sub>1</sub>/D<sub>2</sub> agonist</b> , <b>paraventricular nucleus</b>              | <b>Sublingual 2, 4, 6 mg — inactive if swallowed</b> ; erection in ≈12 min; window ≈2 h | <b>Nausea (commonest)</b> , dizziness, yawning, somnolence, sweating, emesis; <b>syncope</b> with prodrome, no cardiac sequelae; reduced by dose titration | <b>Europe 2001; never US-approved</b>        |
| Melanocortin agonists<br>(melanotan II, PT-141) | Central <b>melanocortin-4 receptor</b> (also food intake, energy expenditure)                    | Early trials effective                                                                  | Flushing, nausea                                                                                                                                           | Not approved for ED                          |
| <b>Trazodone</b>                                | Spinal serotonergic: active metabolite 5-HT <sub>2C</sub> agonist; 5-HT <sub>1A</sub> antagonist | Off-label (priapism association prompted trials)                                        | Drowsiness, nausea, emesis, blood-pressure change, urinary retention, <b>priapism</b>                                                                      | <b>No better than placebo</b> ; limited role |
| L-arginine, L-dopa, limaprost, naltrexone       | NO precursor; dopamine precursor; prostaglandin E <sub>1</sub> ; opioid antagonist               | —                                                                                       | —                                                                                                                                                          | Plausible but insufficiently studied         |

### Intracavernosal Injection

- The **1982 discovery** that intracavernous vasoactive agents induce erection (**Virag**) launched medical therapy for ED.
- Three agents in regular use — **alprostadil, papaverine, phentolamine** — alone or combined (**Bimix, Trimix**). Combinations act **synergistically**, rescue **monotherapy failures** and avoid single-agent side effects (e.g., **alprostadil pain**).
- Start with a **small dose**, especially in **nonvasculogenic ED**. **In-office training** before home use, titrating to an erection rigid enough for intercourse **lasting no more than 1 hour**. Empirical and risk-based dosing give equal satisfaction, outcomes and complications.
- **Contraindications**: psychological instability; history of or risk for **priapism**; **severe coagulopathy**; **unstable CVD**; reduced manual dexterity (a partner may be trained); **monoamine oxidase inhibitors** — an intracavernosal  $\alpha$ -agonist used to reverse priapism could precipitate a **hypertensive crisis**.

### COMPARISON: INTRACAVERNOSAL AGENTS

| Agent                                                 | Mechanism                                                                                           | Pharmacokinetics                                                         | Efficacy                                                | Advantages                                                                          | Disadvantages                                                                                           |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Alprostadil</b><br>(prostaglandin E <sub>1</sub> ) | Receptor → <b>adenylate cyclase</b> → <b>cAMP</b>                                                   | <b>96% metabolised locally within 60 min</b> ; negligible systemic entry | Full erections 70–80% at <b>10–20 <math>\mu</math>g</b> | <b>Only FDA-approved injectable</b> ; least priapism, systemic effects and fibrosis | <b>Painful erection</b> (commonest AE); cost; short shelf-life after reconstitution unless refrigerated |
| <b>Papaverine</b>                                     | <b>Nonspecific PDE inhibitor</b> (cAMP and cGMP) + <b>blocks voltage-dependent calcium channels</b> | Hepatic; half-life 1–2 h                                                 | ≈60%                                                    | Cheap; stable at room temperature                                                   | <b>Liver enzyme rise, priapism, fibrosis</b> → <b>abandoned as monotherapy</b>                          |

| Agent                                   | Mechanism                                                                                                                                                  | Pharmacokinetics | Efficacy            | Advantages              | Disadvantages                                                                      |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-------------------------|------------------------------------------------------------------------------------|
| <b>Phentolamine</b>                     | Postsynaptic $\alpha_1$ block, offset by <b>prejunctional <math>\alpha_2</math> block</b> ( $\uparrow$ noradrenaline)                                      | Half-life 30 min | <b>Poor alone</b>   | Adjunct in combinations | Systemic hypotension, reflex tachycardia, nasal congestion, gastrointestinal upset |
| Vasoactive intestinal polypeptide (VIP) | 28-amino-acid peptide; receptor $\rightarrow$ adenylate cyclase $\rightarrow$ cAMP; once the candidate <b>nonadrenergic noncholinergic (NANC) mediator</b> | —                | Disappointing alone | Works in combination    | With phentolamine (Invicorp) awaiting US approval                                  |

**TABLE 72.8 INTRACAVERNOSAL PHARMACOTHERAPIES**

| Trade name                       | Drug(s)                                 | Concentration                              | Intercourse efficacy             |
|----------------------------------|-----------------------------------------|--------------------------------------------|----------------------------------|
| Caverject; Viridal/Edex          | Alprostadil                             | 5–40 $\mu\text{g/mL}$                      | $\approx 70\%$                   |
| Bimix                            | Alprostadil + phentolamine              | 20 $\mu\text{g/mL}$ + 0.5 mg/mL            | $\approx 90\%$                   |
| Bimix Androskat (European Union) | Papaverine + phentolamine               | 30 mg/mL + 0.5 mg/mL                       | $\approx 90\%$                   |
| <b>Trimix</b>                    | Alprostadil + papaverine + phentolamine | 10 $\mu\text{g/mL}$ + 30 mg/mL + 1.0 mg/mL | <b><math>\approx 90\%</math></b> |
| Invicorp                         | VIP + phentolamine                      | Not available                              | $\approx 80\%$                   |

- **Pattern: combinations ( $\approx 90\%$ ) outperform alprostadil alone ( $\approx 70\%$ );** the term "Bimix" denotes two different pairs.
- **Trap:** alprostadil and VIP act through **cAMP, independent of NO** — which is why ICI works when neurogenic or endothelial failure defeats PDE5Is.

### Intraurethral Suppositories

- Less invasive than injection: drug is absorbed through urethral mucosa into the **corpus spongiosum**, then via small vascular channels into the **corpora cavernosa** — transfer varies with anatomy.
- Early prostaglandin  $E_2$  trial gave full or partial tumescence in a minority; synthetic  $PGE_1$  approved **November 1996** as **MUSE (Medicated Urethral System for Erection)**: a **1  $\times$  3-mm pellet of alprostadil (125, 250, 500, 1000  $\mu\text{g}$ ) deposited 3 cm into the distal urethra**.
- Technique: **insert applicator and deposit pellet  $\rightarrow$  massage to distribute  $\rightarrow$  remain upright for several minutes**; in-office training and dose adjustment first.
- **Responder rate  $\approx 50\%$** ; among responders  $\approx 70\%$  of administrations yield intercourse.
- **ACTIS adjustable constriction band** (FDA-approved) improves retention; **ALIBRA** (alprostadil +  $\alpha_1$ -antagonist **prazosin**) raised home responder rates.
- **Niche role** — inferior to PDE5Is and ICI. Indications: **PDE5I nonresponse from autonomic nerve damage** (radical prostatectomy, cystectomy, trauma); combination with PDE5I; **soft (cold) glans syndrome** (after prosthesis or as an entity).
- Side effects: **urogenital pain (about one-third)**, minor urethral bleeding, hypotension, dizziness; priapism rare.

- Contraindications: alprostadil hypersensitivity, abnormal penile anatomy, priapism-prone conditions. Partner vaginal burning/itching uncommon; **condom mandatory with a pregnant partner**.

### Transdermal/Topical Pharmacotherapy

- Appeal: convenience, simplicity, limited systemic effect; obstacles limit use.
- **Nitroglycerin 2% paste** (NO donor): tumescence but **rarely rigidity; headache in patient and partner** as a systemic vasodilator → abandoned.
- **Papaverine gel**: abandoned — **molecular size too large** for transdermal absorption.
- **Alprostadil with permeation enhancers**, applied intrameatally/to the glans: **Vitaros** (alprostadil 0.3% + proprietary enhancer), **Alprox-TD** (NexACT) — small but significant benefit over placebo; site burning/warmth comparable to placebo.
- **PGE<sub>1</sub> ethyl ester** (prodrug): better permeation, less irritation; applied to the shaft; higher rigidity than placebo. Topical alprostadil will likely share the **role of intraurethral therapy**.
- **MD3000 (Eroxon)**: the **first over-the-counter FDA-approved ED treatment** — glans gel producing **evaporative cooling then gradual warming**, stimulating nerves and cavernosal flow; onset about **10 minutes** in the majority. Vehicle derived from a topical glyceryl trinitrate formulation trial.

### Medical Device

- **Vacuum erection device (VED)**: for men who fail or decline oral or local pharmacotherapy.
- Principle: **negative pressure engorges the penis; a constriction ring retains the blood**. Not a physiological erection — the trapped blood is **predominantly venous** — but sufficient for coitus. The **glans is engorged as well as the corpora** → advantage in **soft glans syndrome**.
- Technique: **clear cylinder over the flaccid penis → manual or battery pump (pressure-release valve prevents injury) → erection → elastic ring rolled to the base → vacuum released and cylinder removed → intercourse; ring off within 30 minutes**.
- **Prescription devices advised; metal or inelastic rings contraindicated**.
- **Efficacy up to 90%** for satisfactory erection, but **satisfaction 30–70%**; long-term continuation up to 60%. Limited in **severe vascular disease** (proximal venous leak, arterial insufficiency) and **fibrosis** after priapism or prosthesis infection.
- **More acceptable to older men in steady relationships** than young single men.
- Counsel on: band discomfort; **pivoting** (rigid only distal to band); **cyanosis and coolness; trapped ejaculate**.
- Complications: minor (pain, numbness, difficult ejaculation, ecchymosis, petechiae); major rare (**skin necrosis, urethral varicosities, Fournier gangrene**). **Caution with anticoagulants (aspirin, warfarin) and bleeding disorders**.
- Special uses: combined with oral, ICI or intraurethral therapy; supplementing a **malfunctioning prosthesis**; preserving tissue elasticity after **priapism, prosthesis explantation or Peyronie surgery**; erectile recovery after prostate cancer treatment.

### Surgery

- Indicated for **penile injury** (genital or pelvic trauma), **structural deformity** (Peyronie disease), **cavernosal fibrosis** after prolonged ischaemic priapism or infection, and when medical therapy is **contraindicated, unsuccessful or undesirable**.

### Penile Prosthesis Surgery

- Creates rigidity by a mechanism unlike physiological or pharmacological erection; **malleable (semirigid)** and **inflatable (hydraulic)** devices; detailed in Chapter 75.

### Penile Revascularization Surgery

- **Arterial revascularisation: inferior epigastric artery** anastomosed to the corpus cavernosum, the **dorsal artery** (revascularisation), the **deep dorsal vein** (arterialisation), or the deep dorsal vein with venous ligation (arterialisation with venous reconstruction).
- Success depends on selection: **arteriography must show the arterial defect** and other organic causes (venous incompetence) must be excluded.
- **Selection criteria:** age under 55, nonsmoker, nondiabetic, no venous leak, radiographic internal pudendal artery stenosis. Best results: **young men (under 30) with isolated stenosis after perineal or pelvic trauma**.
- Complications: **glans hyperaemia (commonest)**, shunt thrombosis, inguinal hernia.
- **Venous reconstruction:** ligation or embolisation of superficial dorsal, deep dorsal or crural veins; crural plication/ligation; pericavernoplasty. Success unconfirmed (diagnosis and correction of the defect imprecise) → **investigational**. Complications: glans hypoaesthesia, skin necrosis, wound infection, curvature/shortening, glans hyperaemia.

### COMPARISON: ARTERIAL VERSUS VENOUS SURGERY

|                        | Arterial revascularisation                               | Venous reconstruction                     |
|------------------------|----------------------------------------------------------|-------------------------------------------|
| Target defect          | Arteriogenic ED (inflow)                                 | Veno-occlusive ED (outflow)               |
| Status                 | <b>Offered to strictly selected men</b>                  | <b>Investigational</b>                    |
| Ideal patient          | <b>Young, post-traumatic, isolated pudendal stenosis</b> | Not defined                               |
| Signature complication | <b>Glans hyperaemia</b>                                  | Glans hypoaesthesia, curvature/shortening |

### Combination Therapies

- Monotherapy nonresponse approaches **40%**; combinations may optimise response or avoid dose-limiting AEs.
- Reported successful pairings: **tadalafil 5 mg daily + sildenafil 50 mg on demand**; PDE5I + psychosocial counselling; PDE5I + T replacement; PDE5I + intraurethral alprostadil; PDE5I + ICI; PDE5I + VED; ICI + VED; intraurethral + VED; intraurethral + penile prosthesis.
- **Caution** for compounded complications; consider **in-office evaluation before home use**.

### Investigational Therapies

- Potential future directions awaiting rigorous safety and efficacy trials.

### Pharmacotherapies

- Repurposed drugs, **off-label and not FDA-approved** for ED.
- **Botulinum neurotoxin (onabotulinumtoxinA) by ICI:** blocks **noradrenaline release from postganglionic sympathetic neurons** so NO-mediated relaxation predominates. Meta-analysis in medication-refractory men and the largest placebo-controlled RCT show improvement through 12 weeks; AEs mild (injection-site pain); **no long-term safety data**.
- **$\beta_3$ -agonist mirabegron:** relaxes corpus cavernosum in human tissue and rats; clinical data conflicting; the only RCT favoured **mirabegron + doxazosin over tolterodine + doxazosin** on IIEF.

### Restorative Therapies

- Proposed for poor PDE5I responders; **expensive**. **Sexual Medicine Society of North America (SMSNA) position:** without regulatory approval, **shock waves, stem cells and platelet-rich plasma are experimental** and require **institutional review board-approved research protocols**.

**COMPARISON: RESTORATIVE AND REGENERATIVE THERAPIES**

| Therapy                                                          | Proposed mechanism                                                                                    | Evidence                                                                                                                                   |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Low-intensity extracorporeal shock wave therapy (Li-ESWT)</b> | Regeneration of <b>nerve, endothelium and smooth muscle</b> (diabetic animal models)                  | Meta-analysis of RCTs in <b>mild-to-moderate ED</b> : improved IIEF erectile function versus sham — <b>strongest evidence of the group</b> |
| Platelet-rich plasma (PRP)                                       | Intracavernosal centrifuged autologous platelets; paracrine/autocrine growth factors                  | Small cohorts positive; <b>RCTs conflicting</b> on clinically important benefit                                                            |
| Stem cells                                                       | Differentiation and tissue replacement (bone marrow, adipose, umbilical cord blood, placental matrix) | Limited <b>uncontrolled</b> human series                                                                                                   |
| Hyperbaric oxygen (HBOT)                                         | 100% oxygen above 1 atmosphere → more dissolved oxygen, tissue delivery, <b>angiogenesis</b>          | <b>Mixed</b> clinical results after positive rodent data                                                                                   |

**Alternative Therapies**

- Herbs, ointments, nutraceuticals and supplements; PDE5I success spawned **counterfeit and imitation products** and unapproved remedies.
- Efficacy of **ginkgo biloba, L-arginine, Korean red ginseng** uncertain without rigorous RCTs; apparent success partly reflects the **large placebo response in ED trials (25–50%)**.
- Mechanism and meaningful efficacy/safety must be shown before advocacy.

**KEY POINTS: TREATMENT CONSIDERATIONS**

- Risk modification — a **healthy, fit lifestyle** — preserves erectile health, though definitive evidence is needed.
- Psychosexual therapy has a role in **integrative management**.
- Oral, intracavernosal, intraurethral and transdermal routes are applied or under study.
- **VED** is an alternative to oral or local pharmacotherapy.
- **Penile prosthesis** when nonsurgical therapy is contraindicated, unsuccessful or undesirable.
- **Arterial revascularisation** only for select men meeting stringent clinical and radiographic criteria.

**Concept** — Every therapy targets one final common pathway, cavernosal smooth muscle relaxation, and the options are ordered by **how much of the man's own physiology they still require**: PDE5Is need intact nerves and endothelium, alprostadil bypasses NO through cAMP, and the vacuum device and prosthesis bypass smooth muscle altogether.

**FUTURE DIRECTIONS**

- Science and technology remain the cornerstones: expect **next-generation pharmacotherapy and gene, stem cell and regenerative therapies**.
- Surgery will evolve from reconstruction and prosthetics to **tissue replacement (penile transplantation)**.
- The paradigm will shift from **symptom- and severity-based** treatment to **cause-specific treatment** guided by disease states and genetics (**precision medicine**).
- The ultimate goal: interventions that **restore and maximise natural erectile function**.